

Varun Prakash J M

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PROFESSIONAL SUMMARY

Dynamic and detail-oriented Software Engineer with a strong foundation in Python, AI/ML, and .NET technologies. Skilled in building and fine-tuning transformer-based models and LLMs, with hands-on experience in vector databases and distributed AI systems. Proficient in cloud platforms (Azure), containerization (Docker, Kubernetes), and scalable microservices development.

PUBLICATIONS

Research Paper

Comparative Study on the Analysis of the Performance of Transfer Learning and the Customized Convolutional Neural Network to Detect Eyewear

IEEE GCAT 2023 (Published October 2023)

Conducted a performance comparison between a custom CNN and transfer learning models for eyewear detection using real-time facial image datasets. Achieved 99% accuracy and demonstrated applicability in security and biometric use cases.

[Read on IEEE Xplore](#)

Blogs

Federated Learning Meets Neural Machine Translation

Explored privacy-preserving multilingual translation for healthcare using federated learning, HuggingFace Transformers, and Optuna. Built a secure client-server model aggregation pipeline using the Flower framework.

[Read on Medium](#)

Automating Dimension Extraction from Engineering Drawings

Presented a system that automates blueprint reviews using PaddleOCR and OpenCV to extract dimensions and reduce manual annotation by over 80%.

[Read on Medium](#)

Automating Meeting Summaries with GPT-4 and LangChain

Built a LangChain-based AI agent that summarizes meeting transcripts and extracts structured key information using GPT-4o-mini. Ensures schema compliance using Pydantic.

[Read on Medium](#)

Building a Helmet Detection System Using Deep Learning and MobileNet SSD

Designed a system to detect helmet usage in real-time using MobileNet SSD and OpenCV. Enabled efficient road safety monitoring through live video analysis.

[Read on Medium](#)

EXPERIENCE

Cognizant Technology Solutions - Programmer Analyst Trainee

November 2024 – Present | Chennai, India

Worked on a large-scale healthcare modernization initiative for major U.S. clients including AllyAlign Health and BayCare Health System. The project focused on upgrading the QNXT claims management system, integrating AI/ML-driven automation, and supporting internal innovation efforts.

Contributions extended across backend system optimization, fraud analytics, federated NLP model development, and blueprint automation—delivering scalable solutions across multiple Cognizant internal use cases.

QNXT Platform Modernization + AI/ML Integration POCs

SQL, PL/SQL, C#, .NET, REST APIs

Led efforts in backend modernization of the QNXT platform while actively contributing to internal Proof of Concepts (POCs) exploring AI/ML-based automation for claims analytics and fraud detection. Focused on building performance-optimized components for real-time claims adjudication, eligibility validation, and provider data orchestration and experimented with ML-based enhancements for long-term innovation in payer-side processing.

Key Contributions:

- Re-engineered SQL/PLSQL queries and developed RESTful APIs in C#/.NET to automate and optimize claims adjudication, member enrollment, and provider search workflows.
- Improved backend performance for real-time eligibility checks and claims processing, ensuring HIPAA-compliant data handling and integration with legacy systems.
- Developed Python-based POCs for fraudulent claim detection using machine learning models (Random Forest, Logistic Regression) and anomaly detection techniques.
- Applied NLP techniques to parse and categorize unstructured provider notes and identify potential denial reasons, contributing to automation and early triaging.

Federated Machine Translation for Healthcare Claims & Communication

PyTorch, Huggingface Transformers, Flower, Optuna

As part of Cognizant's internal innovation labs, designed and implemented a privacy-preserving federated learning pipeline to support multilingual translation of healthcare claims, patient records, and provider communications across distributed client environments. The goal was to explore decentralized AI systems that enable efficient training while maintaining data confidentiality.

Read more in my Medium blog: [Federated Learning Meets Neural Machine Translation](#)

Key Contributions:

- Engineered a federated training system using the Flower framework to support distributed multilingual translation.
- Integrated Optuna for automated tuning of hyperparameters such as learning rate, batch size, and optimizer choice in real-time.
- Designed a secure client-server architecture using model aggregation for privacy-preserving updates.
- Improved training stability and reduced time variance across heterogeneous client devices.

Automated Dimensional Analysis for Engineering Blueprint Review

PaddleOCR, OpenCV, Python

Collaborated on an internal Cognizant R&D project aimed at automating the review of technical engineering drawings for healthcare infrastructure projects. Developed a solution to extract dimensions, annotations, and scale indicators from scanned blueprints to accelerate facility design reviews and reduce manual workload.

Read more in my Medium blog: [Automating Dimension Extraction from Engineering Drawings](#)

Key Contributions:

- Integrated PaddleOCR to identify text and numeric dimension data from noisy image inputs.
- Preprocessed images using OpenCV techniques like skew correction, grayscale conversion, and contour filtering.
- Reduced manual effort in blueprint annotation by over 80% through automated label extraction.

PROJECTS AND RESEARCH

Meeting Summarizer using GPT-4 and LangChain

Personal Project | Python, LangChain, GPT-4o-mini, Pydantic, OpenAI API

Developed an AI-powered agent that automatically summarizes meeting transcripts and extracts key action items and participants using structured language modeling and prompt engineering.

Read more in my Medium blog: [Automating Meeting Summaries with GPT-4 and LangChain](#)

- Key Contributions:**
- Designed an autonomous agent pipeline using LangChain and OpenAI’s GPT-4o-mini for intelligent meeting analysis.
 - Engineered structured prompts to extract topic, summary, tasks, and participants from unstructured meeting text.
 - Implemented Pydantic-based validation to ensure reliable and schema-compliant model outputs.
 - Built a custom tool to persist meeting summaries to local storage using LangChain's Tool abstraction.
 - Achieved consistent parsing and structured output from LLMs, ensuring production readiness of the summarizer.

Helmet Detection using MobileNet SSD Model
Academic Project / Python, OpenCV, MobileNet SSD, Deep Learning

Developed a deep learning-based system to detect motorcycle riders and classify helmet usage from real-time video streams. The model aims to enhance road safety by automating helmet compliance monitoring.

Read more in my Medium blog: [Building a Helmet Detection System Using Deep Learning and MobileNet SSD](#)

- Key Contributions:**
- Implemented MobileNet SSD for fast and accurate object detection in live video feeds.
 - Designed preprocessing pipeline for video frame extraction and real-time analysis.
 - Enabled automatic classification of helmet usage, aiding safety enforcement systems.
 - Achieved reliable detection across diverse lighting and background conditions.

Comparative Study on the Analysis of the Performance of Transfer Learning and the Customized Convolutional Neural Network to Detect Eyewear
Research Project / Python, CNN, Transfer Learning, OpenCV / Published October 2023 (IEEE GCAT 2023)

Conducted a comparative analysis between a custom Convolutional Neural Network (CNN) and transfer learning models for eyewear detection using Python and OpenCV. Evaluated model performance to enhance real-time facial analysis. Published at IEEE GCAT 2023 (October 2023).

Read the published IEEE paper: [IEEE Xplore – GCAT 2023 Paper](#)

- Key Contributions:**
- Designed and trained a custom CNN architecture and compared it with MobileNetV2.
 - Created and labeled dataset with eyewear annotations for supervised learning.
 - Achieved over 99% accuracy in detecting eyewear from face images.
 - Demonstrated effectiveness for security, AR, and biometric use cases.

SKILLS

Languages: Python, C#, SQL, PL/SQL
Frameworks/Tools: PyTorch, Hugging Face Transformers, Optuna, Flower, AutoGen, LangChain
Technologies: Retrieval-Augmented Generation (RAG), Federated Learning, Reinforcement Learning, Large Language Models (LLMs), Natural Language Processing (NLP), Web APIs, Pinecone, Azure OpenAI
Cloud/DevOps: Microsoft Azure, Azure DevOps, Docker, Kubernetes
Databases: SQL, Neo4j

EDUCATION

Bachelor of Engineering (Computer Science & Engineering)	
Global Academy of Technology – Bengaluru	2024
Pre-University Course (PCMB)	
MES PU College of Arts, Commerce & Science – Bengaluru	2018
School	
Cordial High School – Bengaluru	2016

CERTIFICATIONS

- Docker & Kubernetes: The Practical Guide [2025]**
- Azure Machine Learning & MLOps: Benginner to Advance**
- TechA Data Visualization using Tableau Developer Certification**
- Data Science Training**
- SQL for BEGINNERS: Learn SQL using MySQL and Database Design**